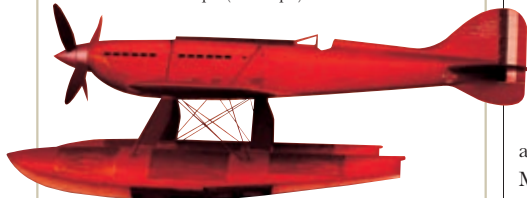


Macchi M.C.72

The beautiful M.C. 72 was built for the 1931 Schneider Trophy but was unable to compete due to the torque effect of its tandem-mounted engine, which made the aircraft uncontrollable on the water. The problem was solved by installing contrarotating propellers to cancel the effect. In October 1934, it set a new world speed record of 440.7mph (709.1kph).



ENGINE 2,800hp Fiat A S 6 tandem V-24 cylinder	
WINGSPAN 31ft 1in (9.5m)	LENGTH 27ft 3in (8.3m)
TOP SPEED 441mph (709kph)	CREW 1
PASSENGERS None	

Supermarine S.6B

The S.6B was the final link in a chain of Supermarine racing seaplanes that had started in 1925 with the unsuccessful S.4. In 1927, the S.5 won in Italy, and in 1929 the S.6 won in England. In 1931, the S.6B completed the course alone, winning the Schneider Trophy outright for Britain after three successive victories. On September 29, the S.6B raised the world "absolute" air speed record to over 400 mph for the first time.



ENGINE 2,350hp Rolls-Royce R V-12 cylinder liquid-cooled	
WINGSPAN 30ft (9.1m)	LENGTH 28ft 10in (8.8m)
TOP SPEED 407mph (655kph)	CREW 1
PASSENGERS None	

Wedell-Williams Model 44

Three Model 44s were built by Jim Wedell between 1930 and 1932. They were among the most consistently successful radial-engined racers of the early 1930s, finishing in the first three in both the Bendix and Thompson Trophy races from 1931 to 1935.

ENGINE 525hp supercharged Pratt & Whitney Wasp Junior	
WINGSPAN 26ft 2in (7.9m)	LENGTH 23ft 4in (7.1m)
TOP SPEED 305mph (491kph)	CREW 1
PASSENGERS None	



One aspect of aircraft performance that especially fascinated the public in the 1920s and 1930s was pure speed. Achievements in this dimension were certainly an impressive token of the general progress of aviation technology. At the Reims meeting in 1909 the top speed of any aircraft had been 48mph (77kph). In 1920 French pilot Sadi Lecoq set a world speed record of 171.1mph (275.2kph) in a Nieuport. In 1928 Italian Major Mario de Bernardi piloted a Macchi M.52 seaplane at 318.6mph (512.7kph). Only three years later, a British Supermarine S.6B seaplane raised the record to 407mph (652kph). Maximum speed had more than doubled in little over a decade, but further advances slowed as designers ran into technological barriers. By 1939, the top speed had progressed to 469.2mph (751.7kph), a record set by German pilot Captain Fritz Wendel in a Messerschmitt Bf209. Much beyond this point piston-engined aircraft could not go. Still, a mere 30 years on from Reims, record speeds had increased almost tenfold.

Popular air races

Not surprisingly in the context of this rapid increase in speeds, air races were as popular as they have ever been. The annual Schneider Trophy seaplane race became a focus for intense national rivalries, drawing substantial investment for engine and airframe development from official sources. Intended by its founder Jacques Schneider as a competition to promote practical reliable seaplanes, it turned into a test bed for single-point designs aimed at speed to the exclusion of all other qualities. Seaplanes were the fastest aircraft of the time, largely because they enjoyed a "runway" of unlimited length for takeoff and landing.

Between 1920 and 1931 the competition was dominated by three companies: Curtiss for the United States, Macchi for Italy, and Supermarine for Britain. The Curtiss R3C, piloted to victory by Doolittle in 1925, was the last biplane to win the Schneider Trophy. After that the Macchi and Supermarine monoplanes predominated, the Supermarines eventually winning the trophy outright for Britain after a "contest" in 1931 in which they were the sole contestants. These light, streamlined metal monoplanes, with their powerful liquid-cooled inline engines, obviously showed the path to the fighter aircraft of World War II, even if the lineage was not as direct as has sometimes been assumed.

In the United States, where private individuals and small companies built and



SCHNEIDER TROPHY

The Schneider Trophy was set up in 1912 by Jacques Schneider – a wealthy industrialist and avid balloon pilot – for a seaplane race of at least 150 nautical miles (173 miles). He hoped the trophy would inspire practical improvements in seaplane design, but it developed into a contest for pure speed.

flew their own aircraft for competitions, air racing was not sharply distinguished from barnstorming. Roscoe Turner, for example, one of the most successful racing pilots, was also a consummate showman, famous for stunts such as flying with a lion cub (wearing a parachute) in the passenger seat. Events such as the National Air Races attracted huge crowds who expected thrills and spills. The top races of the 1930s were the Thompson Trophy, a speed contest flown on a closed course around pylons, and the Bendix Trophy, a race across America from coast to coast. Famous winners included James Doolittle, Roscoe Turner, Jacqueline Cochran, James Wedell, and millionaire Howard Hughes.

Freakish designs

Some custom-built racing aircraft reached extremes of freakish design, none more so than the Gee Bee racers produced by the Granville brothers of Springfield, Massachusetts. These were little more than engines with wings attached, and were regarded as among the most dangerous aircraft ever built. But they were impressively fast. The Gee Bee Model R-1, piloted by Doolittle, won the 1932 Thompson Trophy with a speed of 252.7mph (404.8kph). In a straight line it could reach almost 300mph (480kph). Other victorious aircraft were closer to the mainstream of engine and airframe design: for example, the stylish Northrop Gamma or Howard Hughes' record-breaking H-1 racer. But however much design ingenuity and inventiveness went into speedsters, air racing remained essentially a peripheral if exciting spectacle, not a vital test bed for military or commercial airplane development.

NATIONAL AIR RACE TICKET

This ticket admits one to the 1929 National Air Races, held in Cleveland, Ohio. The race around pylons was won by Douglas Davis in the Travel Air Mystery Ship – giving birth to the annual Thompson Trophy Race.

